

“SMART X-RAY LUGGAGE THREAT DETECTION SYSTEM USING YOLOV8 AND STM32”

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Abstract: This paper presents a smart X-ray luggage threat detection system that integrates a deep learning-based YOLOv8 object detection model with an STM32 microcontroller for automated security response. The proposed system is designed to enhance the efficiency and accuracy of baggage screening processes by reducing human dependency and minimizing manual errors. A custom dataset consisting of X-ray images containing prohibited items such as guns, knives, pliers, scissors, and wrenches is used for training and evaluation. The YOLOv8 model is trained using data augmentation techniques and optimized for real-time inference with an input resolution of 640×640 pixels. The detection results are processed using a decision-making algorithm that classifies the output into SAFE or DANGER states based on confidence thresholds. These results are transmitted via UART communication to the STM32 microcontroller, which controls actuators such as servo motors, LEDs, and buzzer for physical indication and response. Experimental results demonstrate that the system achieves high detection accuracy with low latency, making it suitable for real-time applications. The integration of artificial intelligence with embedded systems provides a cost-effective and scalable solution for modern security checkpoints such as airports, railway stations, and public infrastructure.

Index Terms — YOLOv8, X-ray imaging, object detection, STM32, embedded systems, threat detection

I. INTRODUCTION

The rapid growth in global transportation and increasing security concerns have made baggage screening a critical requirement in public places such as airports, railway stations, and commercial complexes. Traditional X-ray luggage inspection systems rely heavily on human operators, which makes the process time-consuming, inconsistent, and prone to human error. Fatigue, lack of attention, and subjective judgment often reduce the effectiveness of manual inspection systems.

To overcome these limitations, automated threat detection systems based on artificial intelligence and computer vision have gained significant attention. Deep learning models, especially object detection algorithms, have demonstrated exceptional performance in identifying complex patterns in images. Among these, the YOLO (You Only Look Once) family of models is widely recognized for its speed and accuracy in real-time object detection tasks.

In this work, a smart X-ray luggage threat detection system is proposed using the YOLOv8 model integrated with an STM32 microcontroller. The system is capable of detecting prohibited items in X-ray images and triggering automated responses such as gate control, visual indicators, and alert signals. The combination of AI-based detection with embedded hardware control ensures faster decision-making and improved reliability.

The main objective of this study is to develop a low-cost, efficient, and scalable system that can assist security personnel by providing real-time threat detection and automated response mechanisms. The novelty of this work lies in the integration of a real-time YOLOv8-based detection system with an STM32 microcontroller for automated physical response, enabling a complete end-to-end intelligent security solution.

II. RELATED WORK

Significant research has been carried out in the domain of automated threat detection using X-ray imagery. Early approaches primarily relied on traditional image processing techniques such as edge detection, feature extraction, and classification using machine learning algorithms. However, these methods were limited in their ability to handle complex and overlapping objects in X-ray images.

With the advancement of deep learning, convolutional neural networks (CNNs) have become the standard approach for image-based detection tasks. Several studies have explored the use of YOLO-based models such as YOLOv2, YOLOv3, and YOLOv5 for detecting prohibited items in baggage scans. These models provide real-time detection capabilities with improved accuracy compared to conventional techniques.

Recent developments in YOLOv8 have further enhanced detection performance by introducing better feature extraction mechanisms, anchor-free detection, and improved training efficiency. Researchers have demonstrated the effectiveness of YOLOv8 in various domains including medical imaging, surveillance, and security systems.

However, most existing systems focus only on software-based detection and lack integration with embedded hardware for real-time actuation. This creates a gap between detection and response. The proposed system addresses this limitation by integrating the YOLOv8 model with an STM32 microcontroller, enabling automated physical response based on detection results.

III. SYSTEM ARCHITECTURE

A. Block Diagram



The proposed system is designed to automatically detect prohibited items in X-ray luggage images using a combination of deep learning and embedded hardware. The architecture is divided into three major layers: hardware layer, software layer, and AI processing layer. These layers work together to ensure real-time detection and response.

A camera module is used to capture X-ray images of luggage. The captured images are pre-processed and passed to the YOLOv8 object detection model. The model analyses the input image and identifies objects based on trained features, providing bounding boxes and confidence scores for each detected item.

The detection results are processed by a decision-making module, which determines whether the detected object is a threat. If the confidence score exceeds a predefined threshold, the system classifies the output as DANGER; otherwise, it is classified as SAFE. The decision is transmitted to the STM32 microcontroller through UART communication.

The STM32 microcontroller acts as the control unit and executes hardware-level actions such as activating LEDs, triggering a buzzer, and controlling servo motors for gate operation. Additionally, a graphical user interface (GUI) developed using Flask displays real-time detection results and system status.

B. Hardware Layer

The hardware layer consists of an STM32F103C8 microcontroller, servo motors, LEDs, buzzer, and a camera module. The microcontroller controls all output devices based on the received signals.

C. Software Layer

The software layer includes the Flask-based user interface and backend processing system. It handles image acquisition, communication, and visualization of detection results.

D. AI Model Layer

The AI layer is based on the YOLOv8 object detection model trained on X-ray datasets. It is responsible for identifying prohibited objects with high accuracy and speed.

IV. IMPLEMENTATION

The implementation of the proposed system involves both training of the deep learning model and integration with embedded hardware. The system is designed to operate efficiently with minimal latency while maintaining high detection accuracy.

A. Dataset and Training

The YOLOv8 model is trained using a custom dataset containing over 10,000 annotated X-ray images. The dataset includes various classes such as guns, knives, pliers, scissors, and wrenches. Training is performed using an image resolution of 640×640 pixels, a batch size of 16, and 50 epochs. Data augmentation techniques such as horizontal flipping, scaling, translation, and mosaic augmentation are applied to improve model robustness and generalization.

B. Inference Pipeline

During inference, input images are resized and normalized before being passed to the trained YOLOv8 model. The model generates predictions in the form of bounding boxes and class probabilities. These predictions are filtered using a confidence threshold to eliminate false detections. The final output is categorized into SAFE or DANGER states based on detection results.

C. Hardware Integration

The STM32 microcontroller receives UART signals from the processing system, where '0' indicates SAFE and '1' indicates DANGER. Based on the received signal, the microcontroller controls servo motors for gate operation, activates LEDs for visual indication, and triggers a buzzer for alerts. The system is optimized for low latency and reliable communication between software and hardware components.

V. RESULTS & DISCUSSION

A. Performance Metrics

The performance of the proposed system is evaluated using standard metrics such as precision, recall, F1-score, and mean average precision (mAP). The results indicate that the YOLOv8 model achieves high detection accuracy across all object classes. Table 1 shows the performance metrics of the system for different object categories.

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

$$\text{F1-score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

$$\text{mAP} = (1 / N) \times \sum \text{AP}_i$$

(Where AP_i is the average precision of each class and N is the number of classes.)

Table 1: Performance Metrics of Detection Model

Class	Precision	Recall	F1-Score	mAP@50
Gun	0.94	0.87	0.90	0.92
Knife	0.93	0.85	0.89	0.91
Pliers	0.92	0.86	0.89	0.90
Scissors	0.91	0.84	0.87	0.89
Wrench	0.92	0.87	0.89	0.91

These metrics are used to evaluate the detection performance of the proposed system across different object classes.

The average inference time is approximately 150–200 milliseconds per frame, enabling near real-time detection. The integration with STM32 ensures rapid response to detected threats.

Despite promising results, the system has certain limitations such as dependency on controlled lighting conditions and limited field of view due to single camera usage.

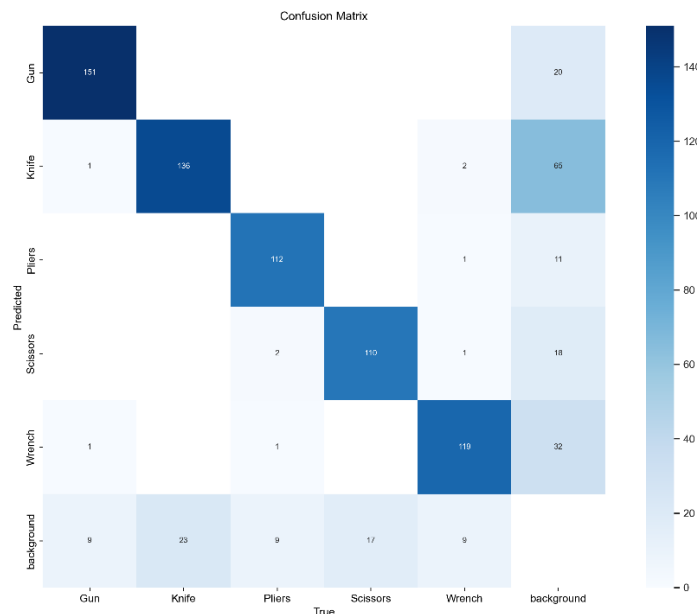


Fig. 1 Confusion matrix representing classification accuracy of the YOLOv8 model.

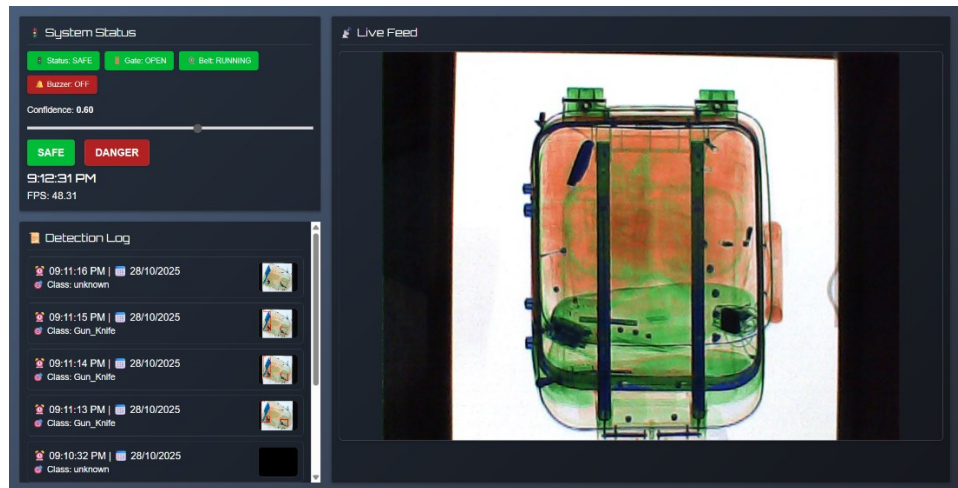


Fig. 2 System dashboard showing SAFE state detection.

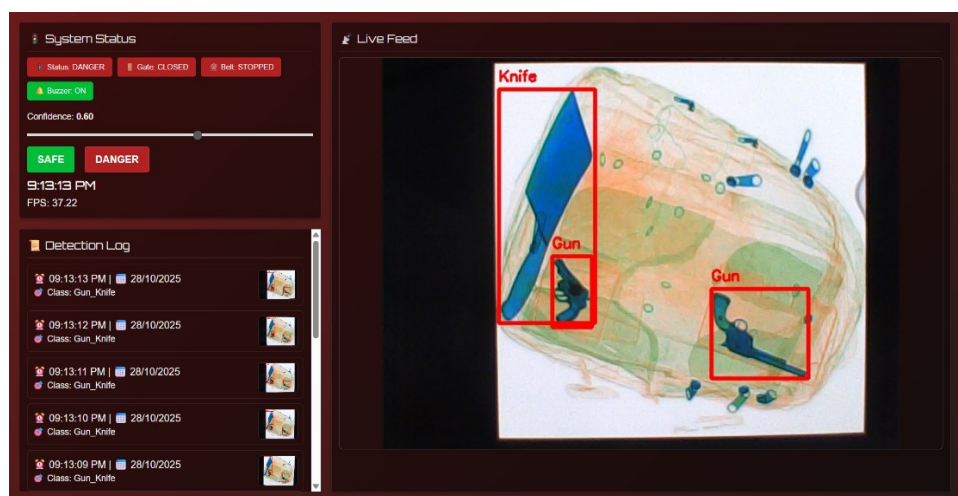


Fig. 3 System dashboard showing DANGER state detection.



Fig. 4 Hardware implementation of the proposed system.

The system performs efficiently under controlled conditions. However, performance may vary with changes in image quality and lighting conditions. Future improvements may include multi-camera setups and deployment on edge AI hardware for enhanced performance.

VI. CONCLUSION

This paper presents a smart and efficient X-ray luggage threat detection system that combines deep learning and embedded systems for automated security applications. The use of the YOLOv8 model enables accurate and real-time detection of prohibited items, while the integration with STM32 ensures immediate physical response.

The proposed system offers a low-cost and scalable solution that can be deployed in various security environments. The results demonstrate that the system is capable of improving detection efficiency and reducing human dependency.

Future work can focus on enhancing system robustness by incorporating larger datasets, multi-camera configurations, and edge computing platforms. The integration of additional sensors and cloud-based monitoring systems can further improve the overall performance and applicability of the system.

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